



JKBOSE 2026

CLASS 11TH

LINEAR INEQUALITIES

MATHS

LECTURE-02

BY-MEHRAJ SIR



TOPICS TO BE COVERED

1

Linear Inequalities

$$ax + b \geq 0, ax + b > 0$$

$$ax + b < 0, ax + b \leq 0.$$

2

Related Problems

EXAMPLE

Solve the inequality: $x^2 - 29x + 100 \geq 0$, where x is a Real number

(Sol) Given that $x^2 - 29x + 100 \geq 0$

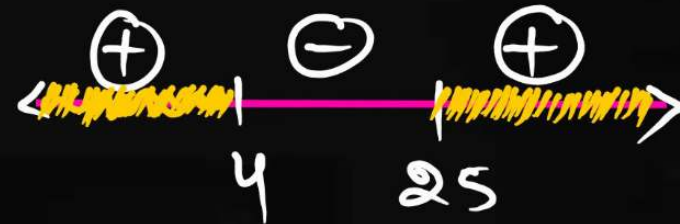
$$\Rightarrow x^2 - 4x - 25x + 100 \geq 0$$

$$\Rightarrow x(x-4) - 25(x-4) \geq 0$$

$$\Rightarrow (x-4)(x-25) \geq 0$$

Therefore by wavy Curve Method, we get;

$$x \in (-\infty, 4] \cup [25, \infty)$$



Therefore the Solution Set = $(-\infty, 4] \cup [25, \infty)$

* We know that the product of two numbers is +ve when the both the numbers are of same sign.

Therefore; we have the following cases:

Case 1st: $x-4 \leq 0$ & $x-25 \leq 0$.

$$\Rightarrow x \leq 4 \text{ & } x \leq 25$$

$$\Rightarrow x \in (-\infty, 4] \text{ & } x \in (-\infty, 25]$$

$$\Rightarrow x \in (-\infty, 4] \cap (-\infty, 25]$$

$$\Rightarrow x \in (-\infty, 4]$$

$$\text{Case 2nd: } x - 4 > 0 \text{ \& } x - 25 > 0$$

$$\Rightarrow x > 4 \text{ \& } x > 25.$$

$$\Rightarrow x \in [4, \infty) \text{ and } x \in [25, \infty)$$

$$\Rightarrow x \in [4, \infty) \cap [25, \infty)$$

$$\Rightarrow x \in [25, \infty)$$

Therefore from both cases, we get;

$$x \in (-\infty, 4] \cup [25, \infty)$$

$$\text{i.e. Solution Set} = (-\infty, 4] \cup [25, \infty)$$

QUESTION

The number of Integers satisfying the inequality $x^4 - 29x^2 + 100 \leq 0$ is

~~1~~

2

~~2~~

4

~~3~~

6

~~4~~

8

(Sol) Given that $x^4 - 29x^2 + 100 \leq 0$.

$$\Rightarrow (x^2)^2 - 29x^2 + 100 \leq 0.$$

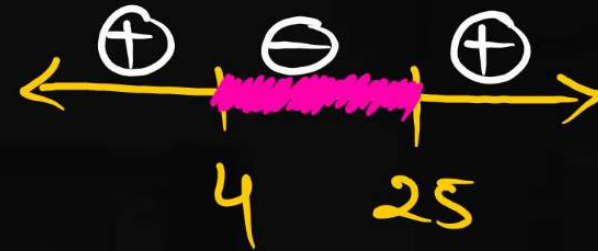
Put $x^2 = t$, we get

$$t^2 - 29t + 100 \leq 0$$

$$\Rightarrow (t - 4)(t - 25) \leq 0$$

By wavy-curve method.

$$t \in [4, 25]$$



$$\Rightarrow x^2 \in [4, 25]$$

$$\text{i.e. } 4 \leq x^2 \leq 25$$

$$\Rightarrow x \in [2, 5] \cup [-5, -2]$$

Therefore the Solution Set = $[2, 5] \cup [-5, -2]$

So, the possible integers are $-5, -4, -3, -2, 2, 3, 4, 5$.

EXAMPLE

The marks obtained by a student of Class XI in first and second terminal examination are 62 and 48 respectively. Find the minimum marks he should get in the annual examination to have an average of at least 60 marks.

(Sol) let the marks scored in 3rd Terminal exam = x .

Then according to the question;

$$\left(\frac{62 + 48 + x}{3} \right) \geq 60$$

$$\Rightarrow \frac{110 + x}{3} \geq 60$$

$$\Rightarrow 110 + x \geq 180$$

$$\Rightarrow x \geq 180 - 110$$

$$\Rightarrow x \geq 70$$

Therefore the required minimum marks = 70.

Hence the result.

EXAMPLE

Find all pairs of consecutive odd natural numbers, both of which are larger than 10, such that their sum is less than 40.

(Sol) let $(x, x+2)$ be the required pair of consecutive odd natural numbers.

Then, according to the Question,

$$x > 10 \quad \text{--- (1)}$$

$$\& \quad x + x + 2 < 40$$

$$\Rightarrow 2x < 40 - 2$$

$$\Rightarrow 2x < 38$$

$$\Rightarrow x < \frac{38}{2}$$

$$\Rightarrow x < 19 \quad \text{--- (2)}$$

Therefore from (1) and (2), we get;

$$10 < x < 19$$

So, the required pairs are $(11, 13)$, $(13, 15)$, $(15, 17)$ & $(17, 19)$

Hence the result.

QUESTION

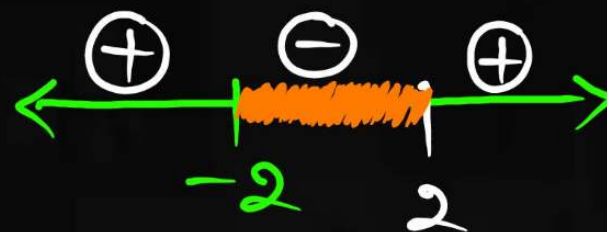
Complete solution of the inequality : $\frac{x-2}{x+2} \leq 0$

- ~~1~~ $[-2, 2]$
- ~~2~~ $(-\infty, -2] \cup [2, \infty)$
- 3 $(-2, 2]$ ✓
- ~~4~~ $[-\infty, -4) \cup [6, \infty)$

(Sol) By wavy-curve Method:

$$x \in (-2, 2]$$

Therefore the solution set = $(-2, 2]$



QUESTION

Complete solution of the inequality : $\frac{x-3}{x+3} \leq 1$

1 $[-3, 3]$ ✗

2 $(-3, \infty)$ ✓

3 $(-3, 3]$ ✗

4 $[-\infty, -4) \cup [6, \infty)$ ✗

(Sol) Given that $\frac{x-3}{x+3} \leq 1$

$$\Rightarrow \frac{x-3}{x+3} - 1 \leq 0$$

$$\Rightarrow \frac{\cancel{x}-3-\cancel{x}-3}{x+3} \leq 0$$

$$\Rightarrow \frac{-6}{x+3} \leq 0$$

$$\Rightarrow \frac{1}{x+3} \geq 0$$

$$\Rightarrow x+3 > 0$$

$$\Rightarrow x > -3$$

So, the solution set = $(-3, \infty)$

QUESTION

Complete solution of the inequality : $\frac{(x+2)(x+3)}{(x-2)(x-3)} \leq 1$

- ☒ 1 $(-\infty, 0]$
- ☒ 2 $(-\infty, 0] \cup (2, 3)$
- ☒ 3 $[2, 3]$
- ☒ 4 $(-\infty, 2) \cup (3, \infty)$

(Sol) Given that $\frac{(x+2)(x+3)}{(x-2)(x-3)} \leq 1$

$$\Rightarrow \frac{(x^2+5x+6)}{(x-2)(x-3)} - 1 \leq 0$$

$$\Rightarrow \frac{(x^2+5x+6) - (x-2)(x-3)}{(x-2)(x-3)} \leq 0$$

$$\Rightarrow \frac{(x^2+5x+6) - (x^2-5x+6)}{(x-2)(x-3)} \leq 0$$

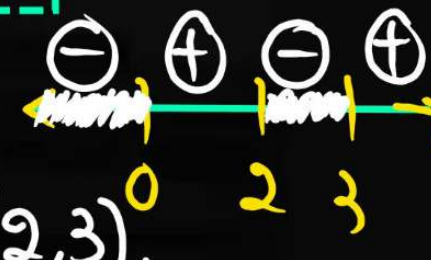
$$\Rightarrow \frac{\cancel{x^2}+5x+\cancel{6} - \cancel{x^2}+5x-\cancel{6}}{(x-2)(x-3)} \leq 0$$

$$\Rightarrow \frac{10x}{(x-2)(x-3)} \leq 0$$

$$\Rightarrow \frac{x}{(x-2)(x-3)} \leq 0$$

Solution Set

$= (-\infty, 0] \cup (2, 3)$



QUESTION

The number of integral values of x if $5x - 1 < (x + 1)^2 < 7x - 3$ is

~~1~~ 0

2 ✓ 1

~~3~~ 2

~~4~~ 3

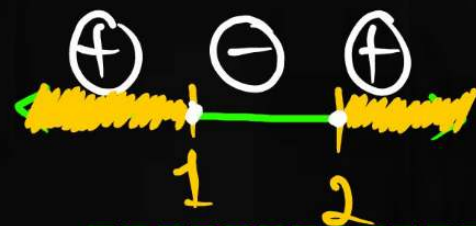
(Sol) Case 1st: $5x - 1 < (x + 1)^2$

$$\Rightarrow 5x - 1 < x^2 + 2x + 1$$

$$\Rightarrow 0 < x^2 + 2x + 1 - 5x + 1$$

$$\Rightarrow x^2 - 3x + 2 > 0$$

$$\Rightarrow (x - 1)(x - 2) > 0$$



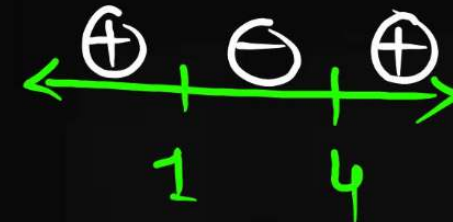
$$\Rightarrow x \in (-\infty, 1) \cup (2, \infty)$$

Case 2nd: $(x + 1)^2 < 7x - 3$

$$\Rightarrow x^2 + 2x + 1 - 7x + 3 < 0$$

$$\Rightarrow x^2 - 5x + 4 < 0$$

$$\Rightarrow (x - 1)(x - 4) < 0$$



$$\Rightarrow x \in [1, 4]$$

Therefore on combining both Cases, we get

$$x \in ((-\infty, 1) \cup (2, \infty)) \cap [1, 4] = (2, 4)$$

Solution Set = $(2, 4)$

$\Rightarrow$ possible integer is 3.

QUESTION

The solution of the inequality : $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$ is

✓ **1** $[1,3] \cup (5, \infty)$

✗ **2** $(1,3) \cup (5, \infty)$

✗ **3** $(-\infty, 1) \cup (5, \infty)$

✗ **4** None of these

(Sol) Given that $\frac{x+7}{x-5} + \frac{3x+1}{2} \geq 0$

$$\Rightarrow \frac{2(x+7) + (3x+1)(x-5)}{2(x-5)} \geq 0$$

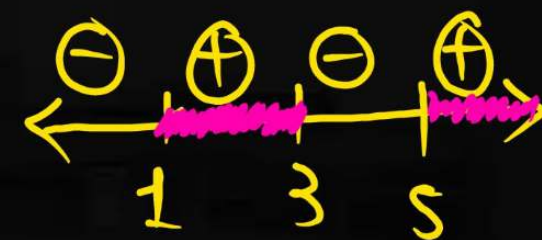
$$\Rightarrow \frac{2x+14+3x^2-15x+x-5}{2(x-5)} \geq 0$$

$$\Rightarrow \frac{3x^2-12x+9}{2(x-5)} \geq 0$$

$$\Rightarrow \frac{3(x^2-4x+3)}{(x-5)} \geq 0$$

$$\Rightarrow \frac{x^2-4x+3}{x-5} \geq 0$$

$$\Rightarrow \frac{(x-1)(x-3)}{(x-5)} \geq 0$$



Solution set = $[1,3] \cup (5, \infty)$.

QUESTION

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

The solution of the inequality : $\frac{|x+2|-x}{x} < 2$ is

1 (0,1)

2 [0,2)

(Homework Problem)

3 $(-\infty, 0) \cup (1, \infty)$

4 None of these

QUESTION

The number of integral values of x satisfying the inequality $\frac{x^2+6x-7}{|x+2||x+3|} < 0$ is

1 5 ✓

2 6 ✗

3 7 ✗

4 8 ✗

(Sol) Given that $\frac{x^2+6x-7}{|x+2||x+3|} < 0$

$$\Rightarrow x^2+6x-7 < 0$$

$$\Rightarrow (x-1)(x+7) < 0$$



So, solution set = $(-7, 1) - \{-2, -3\}$
 $\Rightarrow$ possible integers are $-6, -5, -4, -1, 0$

Thank You

Jan Mehra
1729

Field's Medal
< 40%
after every four years.